

Theoretical Foundations

Below is a brief summary of the knowledge required to solve the problems that follow.

It is by no means a substitute for lectures.

The Concept of a Matrix

If m and n are natural numbers, then a *matrix* A of order $m \times n$ with entries $a_{ij} \in \mathbb{R}$ is defined as

$$A = [a_{ij}]_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}.$$

Formally, a matrix A is a mapping $A : \{1, 2, \dots, m\} \times \{1, 2, \dots, n\} \rightarrow \mathbb{R}$, but we will think of it as a rectangular table of numbers.¹

The vector (or, when appropriate, matrix)

$$[a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}]$$

is called the i -th *row* of matrix A , and we will denote it (internally) by $A_{i \rightarrow}$.

Similarly, the vector (or, when appropriate, matrix)

$$A_{\downarrow j} = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

is called the j -th *column* of matrix A .

When appropriate, we will regard these two objects as matrices of order $1 \times n$ and $m \times 1$, respectively, and when appropriate as vectors from $\mathbb{R}^n$ and $\mathbb{R}^m$.

Among the special cases, we mention the *square matrix*, which has the same number of rows and columns:

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix},$$

as well as the *upper triangular matrix*, which has the form

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}.$$

A *lower triangular matrix* is defined analogously.

A matrix that is simultaneously upper and lower triangular is a matrix of the form

$$\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix},$$

and is called a *diagonal matrix*. The numbers $a_{11}, \dots, a_{nn}$ form its (main) *diagonal*.

Addition, Scalar Multiplication, and Transposition

If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ are matrices and α is a real number, then the sum of matrices A and B is defined as

$$A + B = [a_{ij} + b_{ij}]_{m \times n},$$

that is, every entry of matrix A is added to the corresponding entry (in the same row and column) of matrix B .

Similarly, the scalar product of α and a matrix A is defined as

$$\alpha A = [\alpha a_{ij}]_{m \times n},$$

that is, every entry of matrix A is multiplied by the scalar α .

Transposition is an operation that transforms a matrix of order $m \times n$ into a matrix of order $n \times m$, according to the following rule:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}_{m \times n},$$

$$A^T = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}_{n \times m}.$$

Thus, the rows of matrix A become the columns of matrix² A^T .

Two basic properties of transposition are

$$(A + B)^T = A^T + B^T$$

and

$$(AB)^T = B^T A^T$$

for all matrices A and B of appropriate order (so that the expressions make sense).

¹There are also matrices whose entries are not necessarily real numbers, but, for example, complex numbers, or elements of some *field*, *ring*, etc. The reader is encouraged to learn what a field and a ring are.

²Pronounced “A transposed”.

The Determinant of a Matrix

Every square matrix³ can be associated with a real number $\det(A)$ called the *determinant* of matrix A .

The formal definition of the determinant in terms of permutations of the set $\mathbb{S}_n$ can be found in the textbook, and we will skip it here. Instead, we will give a method for calculating it⁴ using the so-called Laplace⁵ expansion.

Our goal is to determine the determinant of a square matrix $A = [a_{ij}]_{n \times n}$.

First, let us learn how to calculate the determinant⁶ of a matrix of order 2×2 :

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Let us now return to a square matrix $A = [a_{ij}]_{n \times n}$ of arbitrary order n .

We define the *minor* of the entry a_{ij} of matrix A as the determinant obtained from $\det(A)$ by deleting the i -th row and the j -th column, and then calculating the determinant of “what remains”. We denote this minor by M_{ij} .

The *cofactor*⁷ of the entry a_{ij} is defined as

$$A_{ij} = (-1)^{i+j} M_{ij}.$$

It is not necessary, but it is good practice, for the sake of easier tracking, to denote the cofactors using the same letter as the matrix whose cofactors they are.

We are now ready to define⁸ the determinant of matrix A using the *Laplace expansion along the i -th row*:

$$\det(A) = \sum_{j=1}^n a_{ij} A_{ij} = a_{i1} A_{i1} + a_{i2} A_{i2} + \cdots + a_{in} A_{in}.$$

The determinant can also be expanded along a column instead of a row. There is a theorem guaranteeing that the same value is obtained.

Notice that if matrix A is of order n , then its cofactors A_{ij} are determinants of order $n - 1$.

Laplace expansion allows us to reduce a determinant to several determinants of smaller order in this way, and thus recursively until we reach a determinant of order 2, which we know how to calculate.

For a determinant of order three⁹ there is a “trick” for faster calculation, known as *Sarrus’ rule*. The following is an illustration together with the value of the determinant:

³And only such a matrix!

⁴To be precise, if the determinant is defined using permutations, Laplace expansion is a theorem. Equivalently, Laplace expansion can be taken as the definition of the determinant, and then the expression involving permutations can be derived as a theorem.

⁵Pierre-Simon, Marquis de Laplace (1749–1827), French mathematician and astronomer

⁶Square brackets are intended for writing a matrix, while vertical bars are intended for writing a determinant. **These two should not be confused.** A matrix is a **mapping/table** of numbers, while its determinant is a **real number**. Writing a determinant instead of a matrix is not a minor mistake.

⁷A mnemonic: the sign alternates starting with a plus.

⁸Or calculate, if we have defined the determinant using permutations.

⁹**And only for this order!**

¹⁰Nevertheless, we encourage the reader to try to understand them independently before reading about them somewhere or hearing them in lectures, because this will certainly make them easier to remember.

¹¹Here we give an example using the first column of a determinant of order 3; the same holds for any row, column, or any order (dimension) of determinant.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}.$$

In words, the rule is as follows.

First, append the first two columns to the determinant. Then the value of the determinant is obtained by adding the products along all main (“downward”) diagonals, and then subtracting the products along the secondary (“upward”) diagonals.

Properties of the Determinant

Below we list some basic properties of the determinant. We will not explain here why they hold; that is the job of the lectures.¹⁰

1. For every matrix A ,

$$\det(A) = \det(A^T).$$

2. If an entire row or column of a matrix A consists of zeros, then

$$\det(A) = 0.$$

3. Multiplying one row (column) of a determinant by a number and adding it to another row (column) does not change the value of the determinant.

4. If among the rows or columns of matrix A there are two equal or proportional ones, then

$$\det(A) = 0.$$

5. Interchanging two rows (columns) changes only the sign of the determinant.

6. If A is an upper (or lower) triangular matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix},$$

then

$$\det(A) = a_{11}a_{22} \cdots a_{nn}.$$

7. The determinant is additive in any¹¹ row or column, i.e.

$$\begin{vmatrix} a_{11} + \alpha_{11} & a_{12} & a_{13} \\ a_{21} + \alpha_{21} & a_{22} & a_{23} \\ a_{31} + \alpha_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} \alpha_{11} & a_{12} & a_{13} \\ \alpha_{21} & a_{22} & a_{23} \\ \alpha_{31} & a_{32} & a_{33} \end{vmatrix}.$$

8. The determinant is homogeneous in any row or column, i.e.

$$\begin{vmatrix} \alpha a_{11} & a_{12} & a_{13} \\ \alpha a_{21} & a_{22} & a_{23} \\ \alpha a_{31} & a_{32} & a_{33} \end{vmatrix} = \alpha \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

At this point we appeal once again to the reader: **a matrix and a determinant are not the same object.**

A determinant is a number associated with a matrix. Matrices and determinants are not added in the same way and are not multiplied by scalars in the same way, because they are not the same thing.

Matrix Multiplication

So far we have learned how to add two matrices of the same order, as well as how to multiply a matrix by a real number.

Now we will learn how to multiply two matrices.

Let the matrices be

$$A = [a_{ij}]_{m \times k} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{ik} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mk} \end{bmatrix}$$

and

$$B = [b_{ij}]_{k \times n} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{i1} & b_{i2} & \cdots & b_{ij} & \cdots & b_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{k1} & b_{k2} & \cdots & b_{kj} & \cdots & b_{kn} \end{bmatrix}.$$

In order for matrices A and B to be multiplied, the number of columns of matrix A and the number of rows of matrix B must be equal.

In that case, their product is a matrix of order $m \times n$ defined as

$$A \cdot B = \begin{bmatrix} A_{1 \rightarrow} \cdot B_{1 \downarrow} & A_{1 \rightarrow} \cdot B_{2 \downarrow} & \cdots & A_{1 \rightarrow} \cdot B_{n \downarrow} \\ A_{2 \rightarrow} \cdot B_{1 \downarrow} & A_{2 \rightarrow} \cdot B_{2 \downarrow} & \cdots & A_{2 \rightarrow} \cdot B_{n \downarrow} \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ A_{m \rightarrow} \cdot B_{1 \downarrow} & A_{m \rightarrow} \cdot B_{2 \downarrow} & \cdots & A_{m \rightarrow} \cdot B_{n \downarrow} \end{bmatrix},$$

where

$$A_{i \rightarrow} \cdot B_{j \downarrow} = \sum_{t=1}^k a_{it} b_{tj}$$

is the scalar product of the i -th row of matrix A and the j -th column of matrix B .

Properties of Matrix Multiplication, Inverse Matrix

If matrices A and B are also square matrices of the same order, then one very unexpected property holds, known in mathematical literature as the Binet–Cauchy theorem on the multiplicativity of the determinant:

$$\det(A \cdot B) = \det(A) \cdot \det(B).$$

Thus, the determinant of a product is equal to the product of the determinants.

Notice that the products on the left and right sides of the equality are not the same product; they do not even multiply the same “things”.

One important property is that matrix multiplication is an associative operation. However:

MATRIX MULTIPLICATION IS NOT A COMMUTATIVE OPERATION!¹²

The set of square matrices of order n , denoted by $M_n(\mathbb{R})$, together with matrix multiplication can be viewed as the algebraic structure $(M_n(\mathbb{R}), \cdot)$.

For now we know that it has closure¹³ and associativity, but not commutativity.

It is not difficult to verify that the neutral element for matrix multiplication is the so-called *identity matrix* of order n :

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix},$$

so this structure has an identity element.

If, for a matrix $A \in M_n(\mathbb{R})$, there existed a matrix $A^{-1} \in M_n(\mathbb{R})$ such that

$$A \cdot A^{-1} = A^{-1} \cdot A = I_n,$$

then the matrix A^{-1} would be the inverse of matrix A with respect to the operation of matrix multiplication.

This is precisely why it is called the *inverse matrix* of matrix A .

Only square matrices can have inverses, but not every square matrix has an inverse. For example, the first matrix that comes to mind is the zero matrix.

Fortunately, there is a theorem characterizing the invertibility of a square matrix:

A square matrix is invertible if and only if its determinant is nonzero.

¹⁴

It is not difficult to show that for square matrices A and B of the same order,

$$(A^{-1})^T = (A^T)^{-1}$$

and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

¹²This represents one of the more common mistakes in problems involving matrix equations. Be careful!

¹³The product of two square matrices of order n (which is shorthand for $n \times n$) is again a square matrix of order n

¹⁴If we remember that a real number has a multiplicative inverse if and only if it is not equal to zero, then this statement can be expressed more elegantly as follows: **A square matrix has an inverse (in the structure $(M_n(\mathbb{R}), \cdot)$) if and only if its determinant has an inverse (in $(\mathbb{R}, \cdot)$).**

Calculating the Inverse of a Matrix

Since we already knew what the inverse of an element with respect to an operation was, the definition of an inverse matrix was not difficult.

However, we still do not know how to calculate the inverse matrix of a specific matrix¹⁵.

Now we will learn an algorithm for calculating it.

Let

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

be a square matrix, and suppose that we have calculated $\det(A) \neq 0$, so we know that it has an inverse matrix.

For matrix A , its *cofactor matrix* is defined as the matrix obtained from matrix A by replacing each entry with its cofactor:

$$\text{Cof}(A) = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{bmatrix}.$$

The transpose of this matrix is known as the *adjugate*¹⁶ matrix of matrix A :

$$\text{Adj}(A) = \text{Cof}(A)^T = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{bmatrix}^T.$$

Finally, the inverse matrix of matrix A is calculated as

$$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A).$$

Problems Covered in Class

1. Prove that

$$\begin{vmatrix} 2 & -1 & 2 & -1 \\ 0 & 4 & 0 & 1 \\ 2 & 1 & 3 & 1 \\ 2 & 2 & 3 & 2 \end{vmatrix} = 6.$$

Let D be the determinant from the problem.

Using Laplace expansion along the second row¹⁷, we obtain

$$\begin{aligned} D &:= -0 \cdot \begin{vmatrix} -1 & 2 & -1 \\ 1 & 3 & 1 \\ 2 & 3 & 2 \end{vmatrix} + 4 \cdot \begin{vmatrix} 2 & 2 & -1 \\ 2 & 3 & 1 \\ 2 & 3 & 2 \end{vmatrix} \\ &\quad - 0 \cdot \begin{vmatrix} 2 & -1 & -1 \\ 2 & 1 & 1 \\ 2 & 2 & 2 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & -1 & 2 \\ 2 & 1 & 3 \\ 2 & 2 & 3 \end{vmatrix} \\ &= 4 \cdot \underbrace{\begin{vmatrix} 2 & 2 & -1 \\ 2 & 3 & 1 \\ 2 & 3 & 2 \end{vmatrix}}_{D_1} + \underbrace{\begin{vmatrix} 2 & -1 & 2 \\ 2 & 1 & 3 \\ 2 & 2 & 3 \end{vmatrix}}_{D_2}. \end{aligned}$$

Furthermore,

$$\begin{aligned} D_1 &= \begin{vmatrix} 2 & 2 & -1 \\ 2 & 3 & 1 \\ 2 & 3 & 2 \end{vmatrix} = 2 \cdot 3 \cdot 2 + 2 \cdot 1 \cdot 2 + (-1) \cdot 2 \cdot 3 \\ &\quad - (2 \cdot 3 \cdot (-1) + 3 \cdot 1 \cdot 2 + 2 \cdot 2 \cdot 2) \\ &= 12 + 4 - 6 - (-6 + 6 + 8) \\ &= 10 - 8 \\ &= 2 \end{aligned}$$

and

$$\begin{aligned} D_2 &= \begin{vmatrix} 2 & -1 & 2 \\ 2 & 1 & 3 \\ 2 & 2 & 3 \end{vmatrix} = 6 - 6 + 8 - (4 + 12 - 6) \\ &= 8 - 10 \\ &= -2. \end{aligned}$$

Therefore,

$$D = 4D_1 + D_2 = 4 \cdot 2 - 2 = 8 - 2 = 6,$$

which is what was required to prove.

2. Prove:

$$\begin{vmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{vmatrix} = a(b-a)(c-b)(d-c).$$

Using the properties of the determinant, we reduce the determinant to a triangular one:

$$\begin{aligned} \begin{vmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{vmatrix} &= \begin{vmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ 0 & 0 & 0 & d-c \end{vmatrix} \quad IV - III \\ &= \begin{vmatrix} a & a & a & a \\ a & b & b & b \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{vmatrix} \quad III - II \\ &= \begin{vmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{vmatrix} \quad II - I \\ &= a(b-a)(c-b)(d-c), \end{aligned}$$

which is what was required to prove.

3. Determine the inverse matrix of

$$A = \begin{bmatrix} -2 & 0 & 4 \\ 1 & 3 & 1 \\ 1 & 1 & -2 \end{bmatrix}.$$

¹⁵Provided it exists, although this can easily be checked by calculating the determinant.

¹⁶More properly, “adjugate”, but the other form has become established in some literature.

¹⁷It is practical to choose the row/column containing as many zeros as possible, because then fewer 3×3 determinants will have to be calculated.

We proceed according to the algorithm and calculate the cofactors:

$$\begin{aligned} A_{11} &= (-1)^{1+1} \begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix} = -7 & A_{12} &= (-1)^{1+2} \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3 & A_{13} &= (-1)^{1+3} \begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix} = -2 \\ A_{21} &= (-1)^{2+1} \begin{vmatrix} 0 & 4 \\ 1 & -2 \end{vmatrix} = 4 & A_{22} &= (-1)^{2+2} \begin{vmatrix} -2 & 4 \\ 1 & -2 \end{vmatrix} = 0 & A_{23} &= (-1)^{2+3} \begin{vmatrix} -2 & 0 \\ 1 & 1 \end{vmatrix} = 2 \\ A_{31} &= (-1)^{3+1} \begin{vmatrix} 0 & 4 \\ 3 & 1 \end{vmatrix} = -12 & A_{32} &= (-1)^{3+2} \begin{vmatrix} -2 & 4 \\ 1 & 1 \end{vmatrix} = 6 & A_{33} &= (-1)^{3+3} \begin{vmatrix} -2 & 0 \\ 1 & 3 \end{vmatrix} = -6 \end{aligned}$$

Therefore,

$$\text{Cof}(A) = \begin{bmatrix} -7 & 3 & -2 \\ 4 & 0 & 2 \\ -12 & 6 & -6 \end{bmatrix},$$

that is,

$$\text{Adj}(A) = \begin{bmatrix} -7 & 4 & -12 \\ 3 & 0 & 6 \\ -2 & 2 & -6 \end{bmatrix}.$$

Now we calculate the determinant of matrix A :

$$\begin{aligned} \det(A) &= \begin{vmatrix} -2 & 0 & 4 \\ 1 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = \begin{vmatrix} -2 & 0 \\ 1 & 3 \end{vmatrix} - \begin{vmatrix} -2 & 4 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} -2 & 0 \\ 1 & 1 \end{vmatrix} \\ &= 12 + 0 + 4 - (12 - 2 + 0) \\ &= 6. \end{aligned}$$

Finally,¹⁸

$$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A) = \frac{1}{6} \begin{bmatrix} -7 & 4 & -12 \\ 3 & 0 & 6 \\ -2 & 2 & -6 \end{bmatrix}.$$

¹⁸Since most of the second exercise session is spent introducing concepts, explaining their properties, and writing out the procedure for finding an inverse matrix, the few problems solved during the exercises are fairly formulaic. Therefore, this time we will not solve additional problems or provide any for independent work, since that would amount merely to replacing the numbers.